



Paul Perez

Aerospace Engineer in Planetary Sciences

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PROFILE

Aerospace engineer with experience in dispersion modelling and geodata processing. Having worked in the fields of atmospheric forecasting and emergency response scenarios has shown me some of the key environmental and societal challenges we face today. Which is why I now look forward to contributing to Earth monitoring, leveraging geospatial and satellite data into user-focused products for better decision-making.

WORK EXPERIENCE

Royal Military Academy of Belgium (RMA) | Scientific Researcher *Jan. 2024 - Aug. 2025*

- Developed web app for high-resolution near real-time contaminant dispersion forecasting and visualization on interactive map with layers and data insight metrics for enhanced emergency response management.
- Integrated probabilistic prediction approach using forecast ensembles to quantify model uncertainty.
- Studied hypothetical nuclear fallout scenario to explore health impact on neighbouring populations.

Netherlands Institute for Space Research (SRON) | Atmospheric Modelling Intern *Jun. 2020 - Oct. 2020*

- Used HPC system to run ensemble of Weather Research and Forecasting (WRF) model physics configurations for high resolution atmospheric CO₂ transport modelling.
- Quantified uncertainties in CO₂ concentrations due to transport errors in WRF model.
- Identified main physics schemes and emission signals contributing to the uncertainties present.

SAFRAN Transmission Systems | Industrial Management R&T Intern *Apr. 2019 - Aug. 2019*

- Coordinated projects for design, production and testing of metal 3D printed parts to increase MRL level.
- Carried out dimensional repeatability study to evaluate best manufacturing setup and limitations.

EDUCATION

Delft University of Technology (TU Delft) - Netherlands | MSc Aerospace Engineering *2019 - 2023*

- Modules: Physics of the Earth and Atmosphere, Planetary Sciences, Climate Change Science and Ethics, MicroSat Engineering, Numerical Astrodynamics, and more.
- Successfully defended Thesis project about the impact of air pressure on booming sand dunes phenomenon, in the context of Mars exploration and extra-terrestrial soil characterisation.

INSA Lyon - France | BEng Mechanical, Materials and Aerospace Engineering *2016 - 2019*

- Modules: Fluid Mechanics, Aerodynamics, Propulsion Systems, Engineering Mathematics, and more.
- Developed graphical user interface for power transmission software using PyQt for final year project.

SKILLS

- Geospatial Data Processing (GRIB, netCDF)
- Scientific Programming (Julia / Python / Unix shell / Git)
- Applied Research
- Critical Thinking
- Active Listening and Communication
- Open-minded Creativity

LANGUAGES

English Native

French Native

Spanish Upper-Intermediate

ADDITIONAL INFORMATION

- **Certification:** Artificial Intelligence for Earth Monitoring - EUMETSAT
- **Academic Publication:** Perez, P., et al. (2025). Simulation of a nuclear fallout scenario in the Zaporizhzhia nuclear power plant during the Ukraine conflict: Implications of dose rate distributions for downwind populations. *Military Medicine*, 190(Supplement_2), 536–542. <https://doi.org/10.1093/milmed/usaf263>
- **Interests:** Rock Climbing, Musicology, Finding the best coffee in town!